

Power Budget

Team Number: 211

Project Name: EGR 304 Project

Team Member Names: Michael, Levi, Hafsa, Kelton

Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Solenoid	FIT0617	+12 - 24V	1	100	100	mA
	Speaker	50CS08FH-1	+5 - 12V	1	625	625	mA
	MicroController	PIC18F57Q43-CNAI	+1.8-5.5V	1	500	500	mA
	+5V Regulator	LM7805CV	+5-35V	1	1000	1000	mA
	MOSFET	AOTF2618L	(range)	1	1000	1000	mA
<i>+12V Power Rail</i>	Component Name	Part Number	Supply	Qty.	Absolute	Total	Unit
	Solenoid	FIT0617	+12 - 24V	1	100	100	mA
	<i>Subtotal</i>					100	mA
	<i>Safety Margin</i>					25%	
	<i>Total Current Required on +12V Rail</i>					125	mA

+5V Power Rail	Component Name	Part Number	Supply	Qty.	Absolute	Total	Unit
	PSoC™ 4 BLE Module	PIC18F57Q43-CNAI	+1.8-5.5V	1	200	200	mA
	Speaker	50CS08FH-1	+5 - 12V	1	100	100	mA
	MOSFET	AOTF2618L	(range)	0	0	0	mA
						Subtotal	300
						Safety Margin	25%
						Total Current Required on +5V Rail	375
Regulator	+5V Regulator	LM7805CV	(range)	1	1000	1000	

Total Remaining Current Available on +5V Rail						625	mA
External Power Source 1	Component Name	Part Number	Supply	Output	Absolute	Total	Unit
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	12VAC	+V12	5000	5000	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA
	+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA
	+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA
	+5V Regulator	LM7805	+5- 35V	1	1000	1000	mA
Total Remaining Current Available on External Power Source 1						1000	
Calculated Battery Life							
	Component Name	Part Number	Supply		Capacity	Required	
	n/a	n/a	n/a		n/a	#REF!	hours
Battery Life						#VALUE!	
Notes:							